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- Establishing a numerical experimental frame to simulate NMR and two-phase flow
- Implementing NMR diffusion equation in LBM
- Simulate natural complex pore systems

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Joint numerical microscale simulations of multiphase flow and NMR relaxation behavior in porous media using Lattice Boltzmann methods

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Abstract Nuclear magnetic resonance (NMR) relaxometry is a useful tool to estimate transport and storage properties of rocks and soils. However, as there is no unique relation between the NMR signal and these properties in rocks, a variety of empirical models on deriving hydraulic properties from NMR relaxometry data have been published. Complementary to laboratory measurements, this paper introduces a numerical framework to jointly simulate NMR relaxometry experiments and two-phase flow on the micrometer scale. Herein, the NMR diffusion equations were tied to an established Lattice Boltzmann algorithm used in computational fluid dynamics. The numerically simulated NMR data were validated for both **surface-limited and diffusion-limited** relaxation regimes using analytical solutions available for fully and **partially water-saturated** simple pore geometries. Subsequently, simulations were compiled using a complex pore space derived from three-dimensional computer tomography (CT) data of an unconsolidated sand and the results were compared to respective NMR T_1 relaxometry data. The NMR transients simulated for different water saturations matched the measured data regarding initial amplitudes (i.e., porosity and saturation) and relaxation behavior (i.e., distribution of water-saturated pores). Thus, we provide a simulation tool that enables study of the influences of structural and physicochemical properties, such as pore connectivity and pore coupling, surface relaxivity, or diffusivity, on partially saturated porous media, e.g., rocks or soils, with NMR T_1 relaxometry data.

1. Introduction

The application of nuclear magnetic resonance (NMR) to petroleum and environmental research issues has become more important in recent years owing to the method's direct sensitivity to the hydrogen content in pore fluids such as oil, water, or methane gas. The NMR technique is a useful tool that allows inferences of transport and storage parameters such as pore-size distributions or the relative permeability in a porous system, such as rock formations or soils. Even though a considerable amount of data on deriving hydraulic properties of porous media by NMR relaxometry have been published [e.g., Kenyon *et al.*, 1988; Sen *et al.*, 1990; review in Dunn *et al.*, 2002], the majority of these models only empirically address the fully water-saturated state.

The aim of this study is therefore to further advance the fundamental understanding of the relationship between NMR parameters and transport and storage properties of fully and partially saturated pore systems in porous media. To achieve this goal, we numerically simulated NMR relaxometry data and two-phase flow in rock models with complex pore systems derived from micro-CT images. Specifically, Navier-Stokes flow in the partially water-saturated pore space and the evolution of magnetization of the NMR-labeled water molecules governed by diffusion processes were jointly simulated using an adapted Lattice Boltzmann (LB) algorithm. The LB method is a well-established class of tools for computational fluid dynamics (CFD) which can numerically simulate flow in complex geometries. Various model extensions make it possible to also simulate multiphase flow and advection/diffusion processes in porous media [e.g., Rothmann and Keller, 1988; Vogel *et al.*, 2005; Tölke *et al.*, 2006; Ahrenholz *et al.*, 2008]. As discussed in subsequent sections, we employed an extended Rothmann-Keller model coupled with an advection/diffusion model to jointly simulate drainage and imbibition processes along with transient NMR T_1 relaxometry data.

Herein, we (a) simulated the multiphase flow during desaturation and (b) solved the NMR diffusion equation at each saturation level of the system. Subsequently, to validate the model, results of numerical simulations were matched to experimental T_1 NMR relaxometry data [Stingaciu *et al.*, 2010] obtained from a well-sorted unconsolidated sand packing at fully and partially saturated states.

2. NMR Magnetization Evolution in a Porous System

The phenomenon of NMR occurs when the nuclei of certain atoms (e.g., hydrogen) immersed in a static magnetic field B_0 are exposed to a second oscillating magnetic field in the radio frequency range.

The relaxation of observable NMR signals in the pore fluid of a porous system results from the interaction of nuclear spins of the hydrogen protons with their surrounding environment, i.e., with the bulk fluid and interfaces such as pore walls or phase boundaries. This evolution of the magnetization μ during the NMR experiment in a free fluid can be described in terms of the Bloch-Torrey NMR diffusion equation [e.g., Bloch, 1946; Torrey, 1956; Brownstein and Tarr, 1979; Mohnke and Klitzsch, 2010]:

$$\partial \mu / \partial t = \gamma (\mu \times \mathbf{B}^{\text{ex}}) - \begin{pmatrix} \mu_{x'} / T_{2B} \\ \mu_{y'} / T_{2B} \\ (\mu_z - \mu_0) / T_{1B} \end{pmatrix} - \nabla D \nabla \mu, \quad (1)$$

where γ is the gyromagnetic ratio of the protons, $\mathbf{B}^{\text{ex}}$ is the external magnetic field, and T_{1B} and T_{2B} are the fluid's longitudinal and transverse bulk relaxation constants, respectively. The diffusion tensor $\mathbf{D}$ simplifies to $D = \|\mathbf{D}\|$ for the case of isotropic diffusion assumed in this study. Here, we did not consider effects arising from internal magnetic field gradients that may influence the T_2 components of the above equation; thus $T_2 \approx T_1$, and we focused on the longitudinal NMR magnetization behavior characterized by the T_1 relaxation constant.

In the following considerations, the evolution of the longitudinal magnetization μ_z was assumed to commence on terminating the NMR $\pi/2$ -excitation pulse $\mathbf{B}^{\text{ex}}$, i.e., $\frac{\mu_z(t=0)}{\mu_0} = 0$ (and $\frac{\mu_{xy}(t=0)}{\mu_0} = 1$):

$$\partial \mu_z / \partial t = - \frac{\mu_z - \mu_0}{T_{1B}} - \nabla D \nabla \mu_z, \quad (2)$$

where μ_0 is the equilibrium magnetization of the system prior to excitation. When considering multiple NMR active fluids (e.g., oil and water) in a porous system, both, the diffusion constant $\mathbf{D} = \mathbf{D}(\mathbf{r})$ as well as the magnetization at equilibrium $\mu_0 = \mu_0(\mathbf{r})$ will become spatially dependent properties owing to the different types and densities of (immiscible) fluids distributed in the pore space.

Additionally, in a confined system such as a (partially) saturated porous medium, the NMR relaxation rate of excited nuclei can be further accelerated because of surface interactions at the matrix-fluid or fluid-fluid interfaces (e.g., Figure 1a).

In a water-saturated porous medium, the resulting NMR relaxation time relates to the surface-to-volume ratio (s_a/v) of a pore by the surface relaxivity ρ_s (e.g., equation (3)). This surface interaction parameter mainly depends on the presence of ferromagnetic or paramagnetic impurities at the pore surface. Assuming a regime dominated by surface relaxation processes, i.e., the fluid molecules move sufficiently fast and thus explore all parts of the active pore surface s_a with respect to the time scale ($\approx T_1$) of the experiment, the relaxation is monoexponential and the pore size is directly proportional to T_1 , according to

$$\frac{1}{T_1} = \rho_s \frac{s_a}{v} + \frac{1}{T_{1B}}. \quad (3)$$

Upon consideration of a complex porous system, e.g. a porous rock or soil, a multitude of pores within a volume-weighted pore-size distribution (e.g., Figure 1b) will contribute to the resulting NMR T_1 relaxation signal yielding a corresponding NMR relaxation time distribution. The NMR recovery transient is given by

$$e(t) = \sum_i \frac{V_i}{V_0} (1 - \exp(-t/T_{1,i})) \quad (4)$$

with evolution time t , total pore volume V_0 volumetric content v_i , and characteristic relaxation time $T_{1,i}$ of the i -th contributing group of pore sizes in the porous medium [e.g., Kenyon *et al.*, 1989; Dunn *et al.*, 2002].

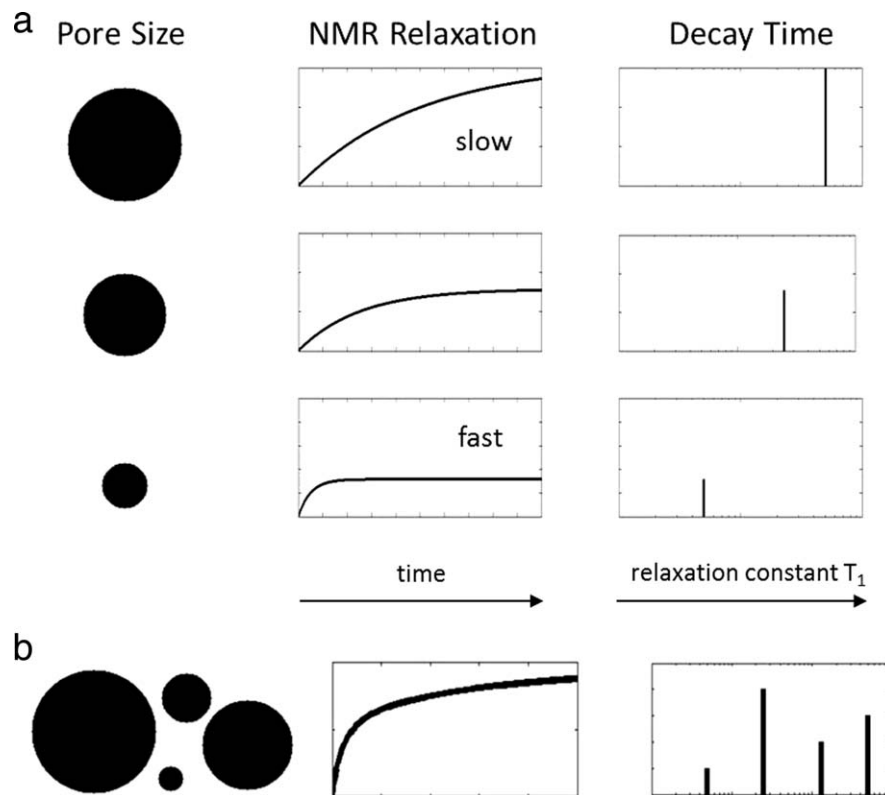


Figure 1. Schematic of the longitudinal NMR exponential relaxation behavior in fully saturated pores related to pore sizes (a) and assuming a uniform surface relaxivity $\rho_s = \text{const.}$ at the pore walls. The relaxation rates ($1/T_1$) increase with decreasing pore sizes. (b) Different pore sizes (pore-size distribution) in a porous system yield a multiexponential NMR decay with a corresponding relaxation time spectrum.

In a diffusion limited regime, e.g. high surface relaxivity, large pore sizes, or a small diffusion coefficient D , the density of unrelaxed spins varies within the pore space. The spins near the surface cannot be replaced immediately after relaxation. Thus, there will be a surplus of unrelaxed spins in the center of a pore compared to the surface region resulting in a multiexponential relaxation characteristic of a single pore [e.g., Brownstein and Tarr, 1979; Mohnke and Klitzsch, 2010].

To account for the accelerated NMR relaxation at the pore walls, we applied Robin boundary conditions, i.e., a linear relationship between the value of the magnetization and its normal derivative. Here, in analogy to a mass transfer coefficient an interface sink term, the surface relaxivity ρ_s , in units of (m/s) is introduced. ρ_s reflects the loss of magnetization at the matrix-fluid and fluid-fluid boundaries, whereby:

$$D(\mathbf{r})\hat{n}_i\nabla\mu(\mathbf{r},t)+\rho_s^i\mu(\mathbf{r},t)|_{S_i^a}=0 \quad \left\{ \begin{array}{l} \rho_s > 0, \text{ active, e.g., pore wall} \\ \rho_s = 0, \text{ passive, e.g., air-water or oil-water} \end{array} \right. \quad (5)$$

with the surface outward normal $\hat{n}_i$ at the i -th solid-water or air-water interface fraction and ρ_s^i is its surface relaxivity. The system's total interface surface area S is given by summing up all active (a) and passive (p) surfaces fractions s of the pore space

$$S=S^{(a)}+S^{(p)}=\sum_i s_i^{(a)}+\sum_i s_i^{(p)}. \quad (6)$$

The total magnetization $M(t)$ of the virtual pore system at a given time t is calculated by integrating

$$M(t)=\int_V \mu(\mathbf{r},t)\partial r. \quad (7)$$

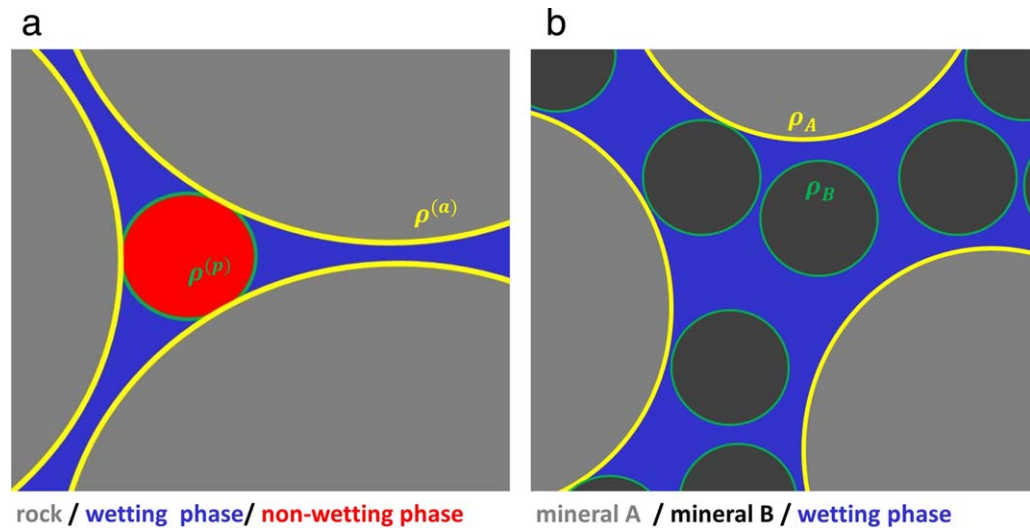


Figure 2. Possible model scenarios for (a) active ($\rho_s^{(a)}$) and passive ($\rho_s^{(p)}$) NMR interfaces of two immiscible wetting and nonwetting fluid phases in a rock's pore space and (b) variations of (active) surface relaxivities attributed to different rock minerals A and B, e.g., $\rho_{sB} \ll \rho_{sA}$ for quartz and magnetite grains, respectively.

Upon considering two immiscible fluids of a partially saturated pore space (cf. Figure 2), e.g., fluid A (wetting) and B (nonwetting), equations (1) and (5) are solved independently for each phase with model properties $\mu_0^{A,B}$, $D^{A,B}$, $s_{(a)}^{A,B}$, and $s_{(p)}^A = s_{(p)}^B$. Consequently, the total magnetization of a numerically simulated NMR relaxometry transient is expressed by

$$M_{\text{tot}}(t) = M_A(t) + M_B(t). \quad (8)$$

Solving the introduced NMR diffusion equations (equation (1)) and boundary conditions (equation (5)) enabled us to comprehensively describe the evolution of magnetization in an arbitrary porous medium, thereby allowing study of a wide variety of solid surface and pore fluid properties that influence NMR relaxometry signatures, e.g., partial saturation, surface relaxivity, bulk relaxation properties, diffusive pore coupling, or slow diffusion processes [e.g., Mohnke and Klitzsch, 2010].

The following section will focus on setting up a numerical framework to jointly simulate two-phase flow and NMR relaxometry experiments for T_1 relaxation processes according to equations (1) and (5).

3. Lattice Boltzmann Method for Modeling Joint Multiphase Flow and NMR Relaxation

Computational fluid dynamics (CFD) have proven to be a viable complementary tool for analyzing and predicting complex transport systems. The LB method is a well-established class of CFD tools that can be used to numerically simulate flow in complex geometries like porous media. Various model extensions allow the additional simulation of multiphase flow and advection/diffusion processes. For this work, an extended Rothmann-Keller model [Rothmann and Keller, 1988] was coupled with an advection/diffusion model using an LB method [Stiebler et al., 2008] to jointly simulate drainage and imbibition processes along with NMR T_1 relaxation. Hereby, we (a) simulated the multiphase flow during drainage or imbibition and (b) solved the NMR diffusion equation at each saturation level of the system.

3.1. Lattice Boltzmann Method for Fluid Flow

The LB method is a numerical method for solving the Navier Stokes equation for incompressible fluids, which, despite the name, is typically used for modelling of low Mach number flow of compressible fluids as well; it is given by the momentum equation

$$\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} = -\frac{1}{\rho} \nabla p + \nu \Delta \vec{u} \quad (9)$$

and the continuity equation

$$\nabla \cdot \vec{u} = 0 \quad (10)$$

for the flow velocity $\vec{u}(t, \vec{x})$ and the pressure $p(t, \vec{x})$ at point $\vec{x}$ and time t , respectively. The LBM is based on a discretization of the Boltzmann equation

$$\frac{\partial f}{\partial t} + \vec{\xi} \cdot \nabla f = \Omega, \quad (11)$$

where $f(t, \vec{x}, \vec{\xi})$ signifies a particle density distribution function over the space of microscopic particle velocities $\vec{\xi}$. Ω models the influence of particle collisions on f . In the limit of small Knudsen numbers and Mach numbers, the solutions of equation (11) can be shown to be solutions of equation (9) as well [Chapman and Cowling, 1970]. The connection between f and the macroscopic variables p and $\vec{u}$ is given by $p = c_s^2 \rho_0 \int f d\vec{\xi}$ and $\rho_0 \vec{u} = \int f \vec{\xi} d\vec{\xi}$, where ρ_0 is a reference density, and c_s is the speed of sound of the medium.

For numerical treatment, equation (11) can be discretized in the velocity space by introducing a finite set of microscopic velocities that satisfy minimal symmetry requirements [Benzi et al., 1992] and, for computational convenience, are arranged to span a cubic lattice. The set used in this work consisted of 19 three-dimensional velocities (D3Q19 in standard terminology [Qian et al., 1992], cf. Figure 3a)

$$\vec{\xi}_q = c \begin{pmatrix} 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix}, \quad (12)$$

where c is a microscopic reference velocity connected to the speed of sound c_s by $3c_s^2 = c^2$. It is noteworthy that the speed of sound is always finite in the LBM, although equation (9) describes fluids in the limit of incompressibility or, equivalently, in the limit of the Mach number $Ma \rightarrow 0$ or the limit of the infinite speed of sound. By treating the fluid as weakly compressible, the LBM introduces a modeling error on the order of Ma^2 , but it gets away without explicitly solving the pressure Poisson equation, which would require large systems of linear equations to be solved.

After discretizing the velocity space, a finite difference discretization for space and time leads to the LB method

$$f_q(\vec{x} + \vec{\xi}_q \Delta t, t + \Delta t) = f_q(\vec{x}, t) + \hat{\Omega}_q(f_0, f_1, \dots, f_{18}), \quad (13)$$

which, as the nodes lie on a Cartesian grid with a grid spacing of $\Delta x = c \Delta t$, can be interpreted as a local collision step $f_q^* = f_q + \bar{\Omega}_q(f_0, f_1, \dots, f_{18})$, $q = 0, 1, \dots, 18$ and propagation of the post collision distributions f_q^* to the neighboring nodes. This locality of the method, in combination with its explicitness, means that it lends itself well to efficient parallelization.

With an appropriate discrete collision operator $\hat{\Omega}$, solutions of equation (13) can still be shown to be solutions of equation (9) in the limit of small Knudsen numbers and Mach numbers, where now $p = c_s^2 \rho_0 \sum_q f_q$ and $\vec{u} = \sum_q f_q \vec{\xi}_{qx}$. The collisions are typically modeled as a relaxation to a Maxwellian equilibrium distribution [Bhatnagar et al., 1954] $\bar{f}^{eq}$ depending only on the local state of the conserved quantities (here, only density and momentum and not energy were considered, as we were not interested in modeling thermal aspects of the flow). Whereas early lattice Boltzmann models used a single-relaxation time for all distributions, the collision operator used in this work is a multirelaxation time model [Lallemand and Luo, 2000]

$$\hat{\Omega} = \mathbf{M}^{-1} \mathbf{S} (\mathbf{M} \mathbf{f} - \mathbf{m}^{eq}), \quad (14)$$

where the matrix $\mathbf{M}$ transforms the set of distributions into physically meaningful moments and $\mathbf{m}^{eq} = \mathbf{M} \mathbf{f}^{eq}$ is the discrete analogue of a Maxwellian equilibrium. $\mathbf{S}$ is a diagonal matrix containing specific relaxation rates for each of the moments, which can be tuned to optimize the stability and accuracy of the numerical method. The version used in this work was optimized for Stokes flow and was first presented in Tölke et al. [2006]. Details about the collision operator (A and B) and the extensions for multiphase flow (C) can be found in the Appendices A, B, and C.

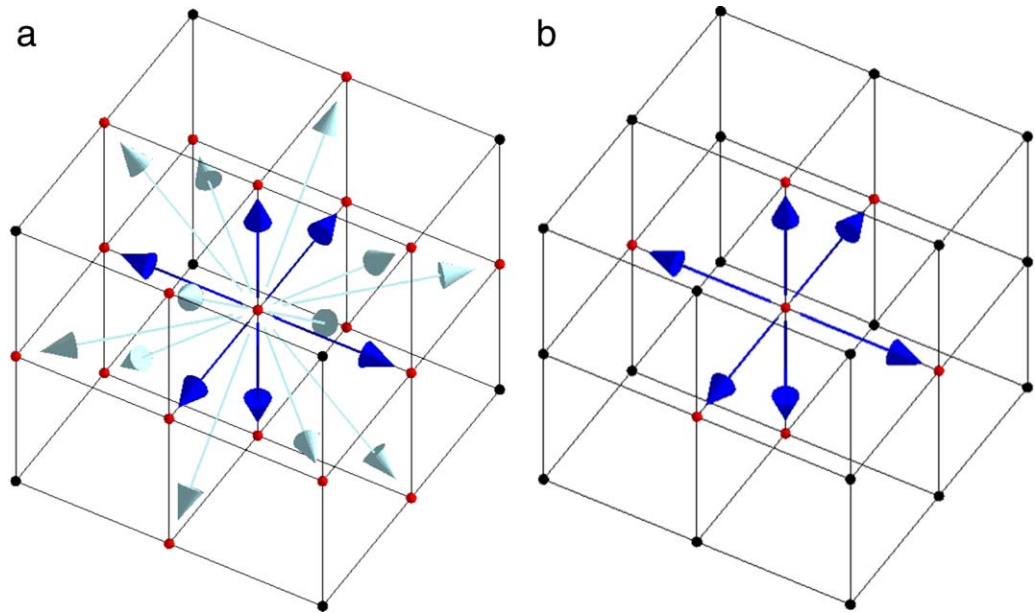


Figure 3. In the D3Q19 model (a), each node on the three-dimensional Cartesian grid is connected to 19 (red) of its 27 neighbors (including the node itself). As the grid spacing is $\Delta x = c\Delta t$, density distribution functions are propagated exactly to one of those neighbors during one time step. The set of microscopic velocities contains members with absolute values of 0 (invisible, connecting to the node itself), c (dark blue), and $\sqrt{2}c$ (light blue). In the D3Q7 model (b), each node on the three-dimensional Cartesian grid is connected to 7 (red) of its 27 neighbors (including the node itself). The set of microscopic velocities contains members with absolute values of 0 (invisible, connecting to the node itself) and c (dark blue).

3.2. LB Model for NMR Diffusion

By choosing a different collision operator, the LB formalism can also be applied for solving other types of conservation-based physical models, such as diffusion equations. Like the original LB method for fluid flow, these methods are well suited for parallelization and for simulating processes in complex geometries. Moreover, they are advantageous for coupled flow problems where an LB solver has already been used for the flow; this is because the numerical grid is usually the same as the one used for the flow problem. Thus, coupling tends to be straightforward.

The LB model we chose for solving the NMR diffusion equation (equation (2)) is a member of a family of advection/diffusion LB algorithms described in *Ginzburg* [2005a]. The particular member we chose is an equilibrium type model, i.e., the diffusivity of the fluid is set via a parameter in the quasi-Maxwellian equilibrium distribution rather than via the relaxation rate of the distributions, which is fixed at a value that optimizes the numerical properties of the model. As diffusion is a structurally simpler problem than the Navier-Stokes equation, a smaller set of macroscopic velocities suffices for its solution. In order to save memory and floating point operations, we chose a set of only seven microscopic velocities (D3Q7, Figure 3b), although the isotropy properties of such a method are compromised. Accordingly, we introduce a set of seven distributions as primary variables, which we call g_q to distinguish them from the distributions for the Navier-Stokes equations. The distributions are connected to the longitudinal magnetization via

$$\mu_z = \sum_{q=0}^{q=6} g_q = \sum_{q=0}^{q=6} g_q^{\text{eq}}. \quad (15)$$

The method is given by the following evolution equation:

$$g_q(\vec{x} + \Delta t \vec{\xi}_q, t + \Delta t) = g_q(\vec{x}, t) + \lambda [g_q(\vec{x}, t) - g_q^{\text{eq}}(\vec{x}, t)] - s_q \Delta t \frac{\mu_z - \mu_0}{T_{1B}} \quad (16)$$

with the set of microscopic velocities given by

$$\vec{\xi}_q = c \begin{pmatrix} 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}. \quad (17)$$

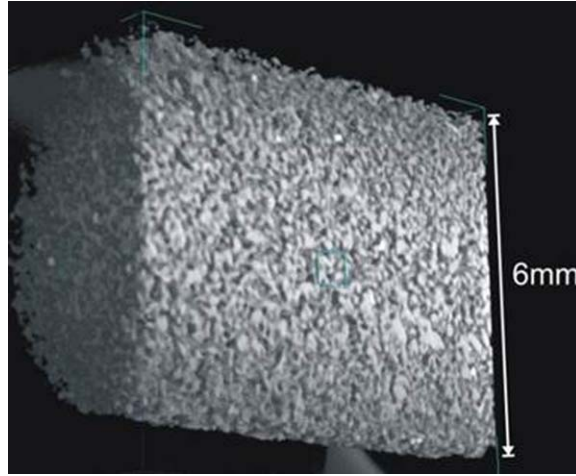


Figure 4. Micro-CT scan of an unconsolidated medium sand sample (FH31) (voxel size 15 μm) with pore sizes of 40–300 μm .

The equilibrium distributions are given by

$$g_q^{\text{eq}} = \mu_z \left(t_q + \frac{1}{3} a^{(e)} t_q^{(e)} \right) \quad (18)$$

with μ_z given by equation (15) and the relaxation rate λ set to

$$\lambda = -3 + \sqrt{3}. \quad (19)$$

s_q and $t_q^{(e)}$ are vectors that ensure consistent discretization of the source term and the diffusive term in equation (2), respectively, and which can be obtained from second-order polynomials in $\vec{\xi}_q$ as derived in [Ginzburg, 2005a]:

$$s_0 = t_0 = 0, s_q = t_q = \frac{1}{6} \text{ for } q \neq 0 \quad (20)$$

and

$$t_0^{(e)} = -3, t_q^{(e)} = \frac{1}{2} \text{ for } q \neq 0. \quad (21)$$

The diffusion coefficient of the model is set via the parameter $a^{(e)}$:

$$a^{(e)} = \frac{6\sqrt{3}D}{c^2\Delta t} - 1. \quad (22)$$

Following Ginzburg [2005b], the Robin boundary conditions for the NMR diffusion equation (equation (5)) are implemented as a modified bounce-back scheme

$$g_{\bar{q}}(\vec{x}_B, t + \Delta t) = g_q^*(\vec{x}_B, t) - (\vec{n} \cdot \vec{\xi}_i) \rho_s, \quad (23)$$

where g^* is the prepropagation state of g as given by the right-hand side of equation (16),

$$g_q^*(\vec{x}, t) = g_q(\vec{x}, t) + \lambda \left[g_q(\vec{x}, t) - g_q^{\text{eq}}(\vec{x}, t) \right] - s_q \Delta t \frac{\mu_z - \mu_0}{T_{1B}}. \quad (24)$$

This scheme replaces the normal propagation at boundary nodes (fluid nodes that have at least one solid neighbor located at $\vec{x}_B$) for all q such that $(\vec{x}_B + \vec{\xi}_q \Delta t)$ is the location of a solid node. $\bar{q}$ is related to q via $\vec{\xi}_{\bar{q}} = -\vec{\xi}_q$.

Consequently, for the boundary conditions on passive interfaces, i.e., surface relaxivity $\rho_s = 0$, between different fluid phases, a simple halfway bounce back was used:

$$g_{\bar{q}}(\vec{x}_B, t + \Delta t) = g_q^*(\vec{x}_B, t). \quad (25)$$

4. Model Validation With Analytical Solutions for Spheres

In order to quantitatively characterize complex 3-D pore systems (e.g., Figure 4) with NMR relaxometry, one typically reverts to pore volume equivalent simple geometrical models, e.g., sphere ensembles or capillary bundles [e.g., Kenyon *et al.*, 1989].

For such geometries, Brownstein and Tarr [1979] introduced analytical solutions of the Bloch-Torrey equations—valid for fast and slow diffusion regimes—in terms of their respective eigenvalues for cylindrical, spherical, or planar pore shapes.

To validate the introduced LB implementation of the NMR diffusion equations, three different principle model setups using spherical pore geometries were simulated and compared to the respective analytical

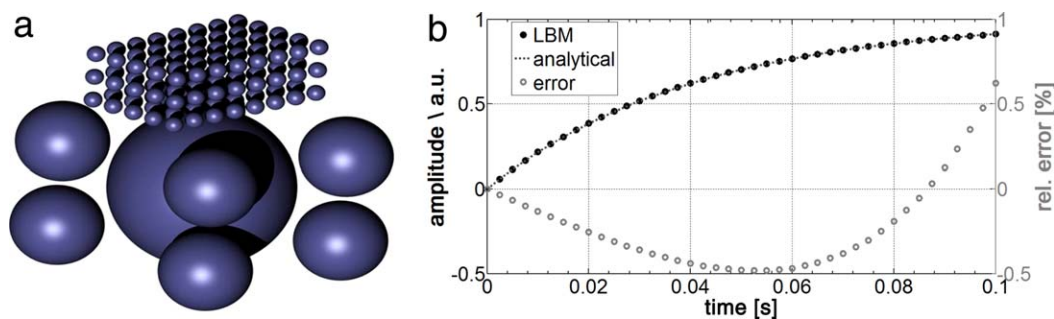


Figure 5. (a) Cluster of isolated spherical pores ($R=10$, 5, and 2 μm). (b) Simulated T_1 relaxation data ($\rho=10\text{ }\mu\text{ms}^{-1}$; $D=2\cdot 10^{-9}\text{ m}^2\text{s}^{-1}$), analytical solutions and fit errors. The NMR amplitudes are given in arbitrary units (a.u.). The deviation (relative error) between numerical and analytical solutions is given in %.

solutions. The validation scenarios comprised fast and slow diffusion environments and partially saturated pores with NMR signals originating from a single (water) and both fluid phases (water and oil).

Shown in Figure 5a is the superposition of simulated T_1 relaxation transients calculated for three fully saturated isolated spherical pore clusters (Figure 5b) with respective pore radii R of 10, 5, and 2 μm . The LB grid was chosen to satisfy a node distance of $\Delta r \leq R/50$. Simulations were calculated for a surface relaxation-dominated relaxation regime (fast diffusion), i.e., independently of its shape, a pore is characterized by a single-relaxation time. Here, the NMR parameters included a surface relaxivity of $\rho_s=10\text{ }\mu\text{m/s}$ and a diffusion constant of $D=2.0\cdot 10^{-9}\text{ m}^2\text{s}^{-1}$, thus $\frac{\rho_s R}{D} \ll 1$. Bulk relaxation time of water was assumed as $T_{1B}=2\text{ s}$. Simulations of NMR relaxation signals (normalized amplitudes) arising in a diffusion-dominated relaxation regime (slow diffusion) within a spherical pore, i.e., $\frac{\rho_s R}{D} \geq 1$, are shown in Figure 6. Here, as a consequence of a diffusion-dominated regime (slow diffusion), multiexponential relaxation occurred for a single (spherical) pore, both for a radius of 50 μm and a radius of 100 μm . The simulations were generally in good agreement with the respective analytical solutions both for fast and slow diffusion models. The observed amplitude misfit of $<1\%$ was attributed to discretization errors as a consequence of the chosen grid.

For consideration of two-phase systems, isolated partially water-saturated pores containing a concentric spherical entrapment (cf. inset of schematic view in Figure 7a) of the nonwetting fluid, e.g., air, were simulated and matched to transient data obtained from analytical solutions [Brownstein and Tarr, 1979; Costabel, 2011].

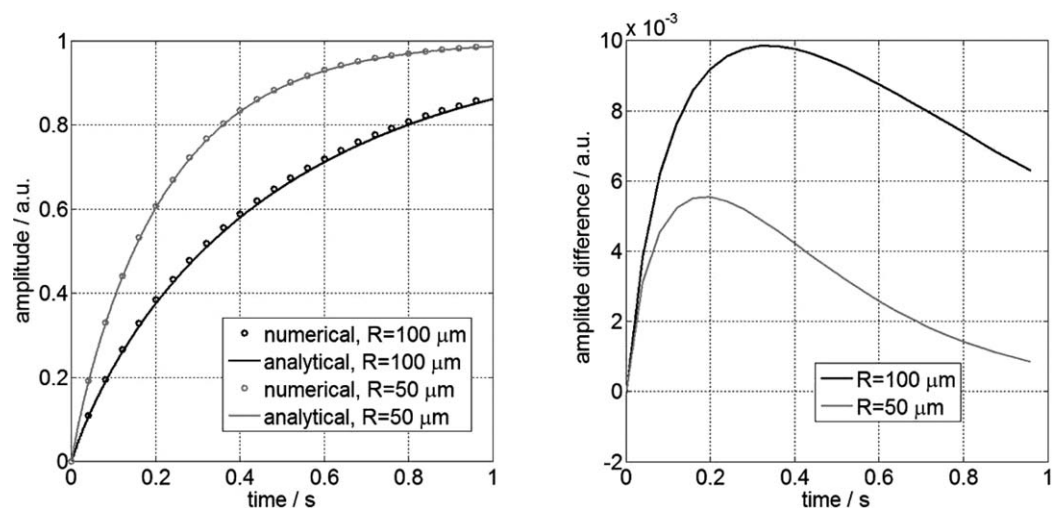


Figure 6. Lattice Boltzmann NMR simulations for two spherical pores ($R_1=50\text{ }\mu\text{m}$, $R_2=100\text{ }\mu\text{m}$) in slow diffusion environments with relevant NMR parameters of the fluid $D=2\cdot 10^{-9}\text{ m}^2\text{s}^{-1}$, $T_b=2\text{ s}$, and the pore's surface relaxivity $\rho_s=100\text{ }\mu\text{ms}^{-1}$. (left) NMR T_1 signal. (right) Deviation of the simulated NMR response from the analytical solution. The NMR amplitudes and respective mismatches are given in arbitrary units (a.u.).

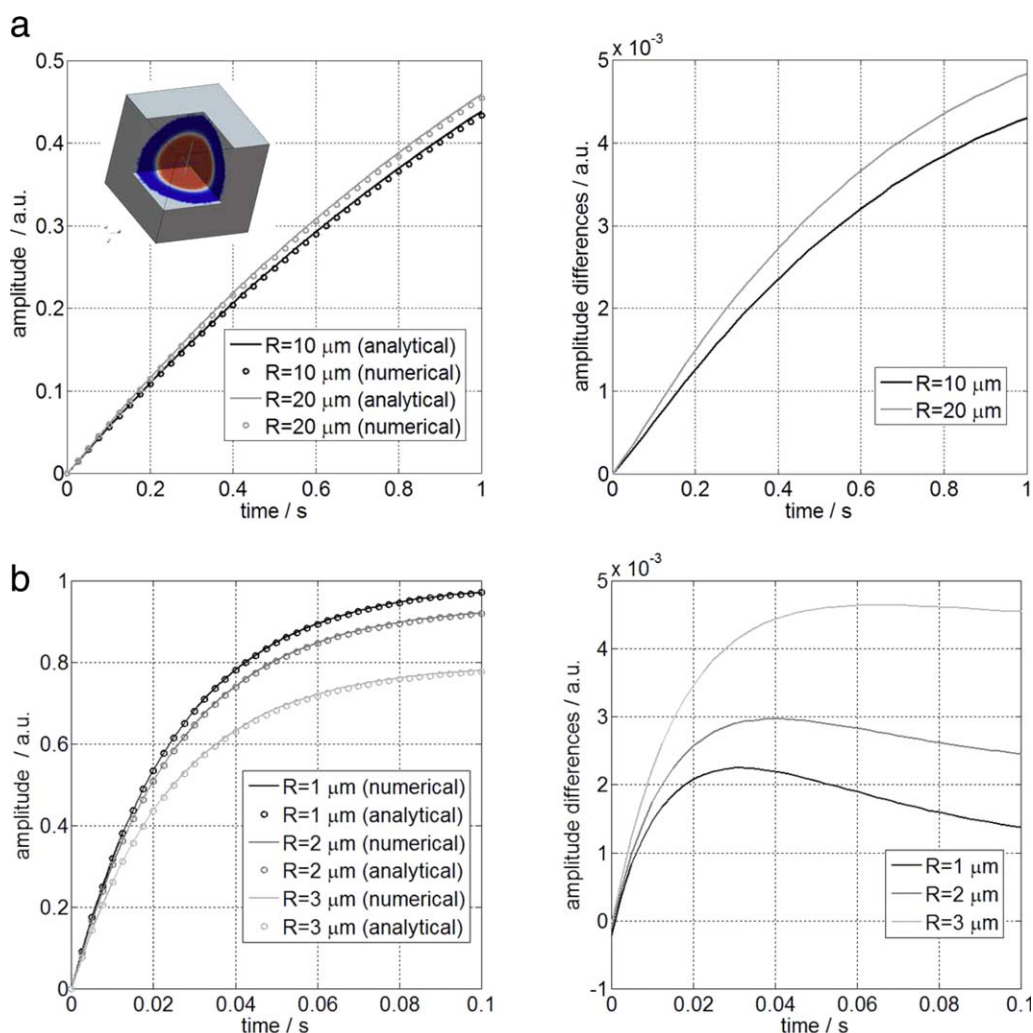


Figure 7. Comparison of numerical and analytical NMR T_1 relaxation signals of a partially saturated spherical pore with spherical entrapment of nonwetting fluid (red) and wetting fluid (blue) phases (inset figure in top left corner). The left figures show the numerical and analytical NMR signals and the right figures the mismatch between both signals (a) for a spherical air entrapment (radii 10 μm and 20 μm) in a pore with a radius of 50 μm and (b) for a spherical water entrapment (radii 1, 2, and 3 μm) in an oil-wet pore with a radius of 5 μm . The NMR fluid parameters used for calculation of analytical and simulated NMR relaxation (T_1) data of water and oil phases are given by $D_{\text{water}} = 2 \cdot 10^{-9} \text{ m}^2 \text{ s}^{-1}$ and $D_{\text{oil}} = 5 \cdot 10^{-11} \text{ m}^2 \text{ s}^{-1}$ (diffusion constants) and $T_{b,\text{water}} = 2 \text{ s}$ and $T_{b,\text{oil}} = 0.03 \text{ s}$ (bulk relaxation time). NMR amplitudes and respective mismatches are given in arbitrary units (a.u.).

Figure 7a shows the simulation results of NMR transients for an air bubble trapped in a partially water-saturated spherical pore. The air-water interface was considered to be a passive boundary ($\rho_s = 0 \text{ } \mu\text{m/s}$) and the water-rock interface an active boundary ($\rho_s = 10 \text{ } \mu\text{m/s}$). Simulation results for the converse case of an oil-wet scenario, i.e., water ($D_{\text{water}} = 2 \cdot 10^{-9} \text{ m}^2 \text{ s}^{-1}$) entrapment in an oil ($D_{\text{oil}} = 5 \cdot 10^{-11} \text{ m}^2 \text{ s}^{-1}$) filled spherical pore, which are shown in Figure 7b, were matched with the respective analytical solutions. Here, the NMR signal arising from the trapped water (bulk relaxation time $T_{1B,\text{water}} = 2 \text{ s}$) superimposes to the relaxation signals of the adsorbed oil phase ($T_{b,\text{oil}} = 0.03 \text{ s}$), i.e., $m(t) = m_w(t) + m_o(t)$.

The LB simulation results were in very good agreement with the analytical models showing deviations of $< 0.1\%$ which could be further reduced by increasing the number of nodes of the LB grid.

In conclusion, the simulated NMR relaxation signals for a single fluid and for two immiscible fluid phases as obtained by the implementation of the NMR T_1 diffusion equation in the adapted LB model framework were in excellent agreement with results calculated by the respective analytical NMR models, both for surface and diffusion limited relaxation environments. In the subsequent section, we compare experimental and modeled data on the basis of a computer tomography (CT) image of an unconsolidated packed sand sample (FH31) with the complex geometry encountered in a natural pore space as shown in Figure 4.

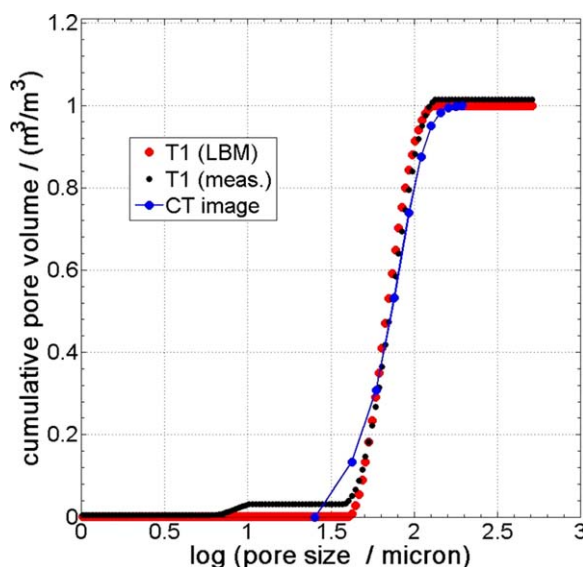


Figure 8. Cumulative pore-size distributions for the FH31 sand sample as derived from the 3-D CT image (blue), simulated T_1 data using LB simulations (red), and T_1 measurements (black dots) on the fully saturated sample. Contributions of small pore sizes (~ 10 – $30 \mu\text{m}$) in measured data reflect the inverted short T_1 components.

5. Results

Joint numerical two-phase flow (drainage) and NMR saturation recovery experiments were simulated on the basis of a 3-D micro-CT image (voxel size $15 \mu\text{m}$) of a packed medium sand sample (FH31) with grain sizes between 0.18 and 0.72 mm which was experimentally investigated by Pohlmeier *et al.* [2009] and [Stingaciu *et al.*, 2010] using multistep-outflow, mercury injection (HgPor), and T_1 NMR relaxometry experiments. Based on these experimental results, the (active) surface relaxivity of the LB model space and parameters were calibrated to $\rho_s = 12 \mu\text{m/s}$, porosity ($\phi = 0.36$) and specific surface ($S_{\text{por}} = 2.4 \times 10^5 \text{ m}^{-1}$).

The pore-size distribution presented in Figure 8 was deduced from the micro-CT image (cf. Figure 4) after tri-

angulation with a marching cube algorithm [Ahrenholz *et al.*, 2008] by means of 3-D image analysis, e.g., Paraview.

All LB simulations introduced in this section were performed on a numerical domain of $200 \times 200 \times 240$ grid nodes, which, with a grid spacing of $15 \mu\text{m}$ correspond to a real domain size of $3 \times 3 \times 3.6 \text{ mm}^3$. Only $32 \cdot 10^6$ of these nodes were active, whereas the rest of the nodes represented the solid matrix. The simulations were run on a computer cluster using 36 CPU nodes (288 cores) with 2.6 GHz Intel Quad X5550 processors. The simulation code was implemented in Fortran 90 and parallelized using Message Passing Interface (MPI). With that setup, $\sim 1.4 \times 10^8$ node updates per second (wall time) could be achieved, or ~ 4.4 numerical time steps per second.

We performed a numerical drainage experiment by gradually increasing pressure at the lower boundary over the course of 1.8×10^6 numerical time steps (without including gravity) resulting in a correspond-

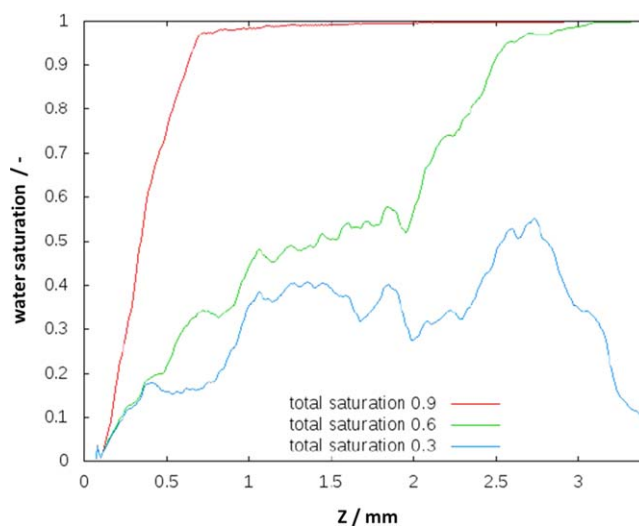


Figure 9. Saturation over sample height for different total saturation levels during the numerical drainage experiment.

ing pressure drop across the sample length with its “open” boundary at $z = 3.6 \text{ mm}$. The evolution of the saturation distribution over the height of the sample is shown for three saturations in Figure 9. Whereas at 90% a strong saturation gradient for $z < 0.5 \text{ mm}$ (inflow at $z = 0 \text{ mm}$) could be observed, the saturation profile became more flat at lower saturations, i.e., after breakthrough of air had occurred. This behavior is supported by observations made by Green *et al.* [2007] who studied saturation profiles of centrifuge capillary pressure measurements using magnetic resonance imaging (MRI). Similar to the author’s

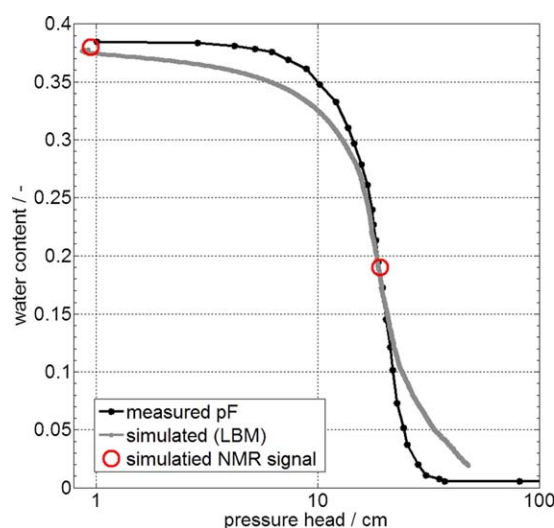


Figure 10. Measured water retention curves [Stingaciu et al., 2010] and LB simulations based on a 3-D micro-CT image for the FH31 sand sample. Circles mark saturation states used for LB simulation of NMR T_1 relaxation data.

laboratory observations the profiles shown in Figure 9 also reflect the capillary pressure gradient and corresponding water content variations along the desaturated porous system. This qualitative similarity indicates the model approximates a representative elementary volume (rev). Analogous to centrifuge measurements, a more constant saturation profile along the sample may be achieved by switching the inflow and outflow and compiling a second drainage run, i.e., inflow at $z=3.6$ mm. As a consequence of this heterogeneous water distribution in the pore space also the peak position of the relaxation time distribution ($T_1 \propto R$) remains at $T_{\text{peak}}=1.2$ s until a significant fraction of the water sitting in the larger pores is displaced throughout the sample.

Figure 10 shows water retention curves resulting from LB simulations and respective pressure-saturation data obtained from laboratory measurements [Stingaciu et al., 2010]. The simulated data show a more gradual decrease

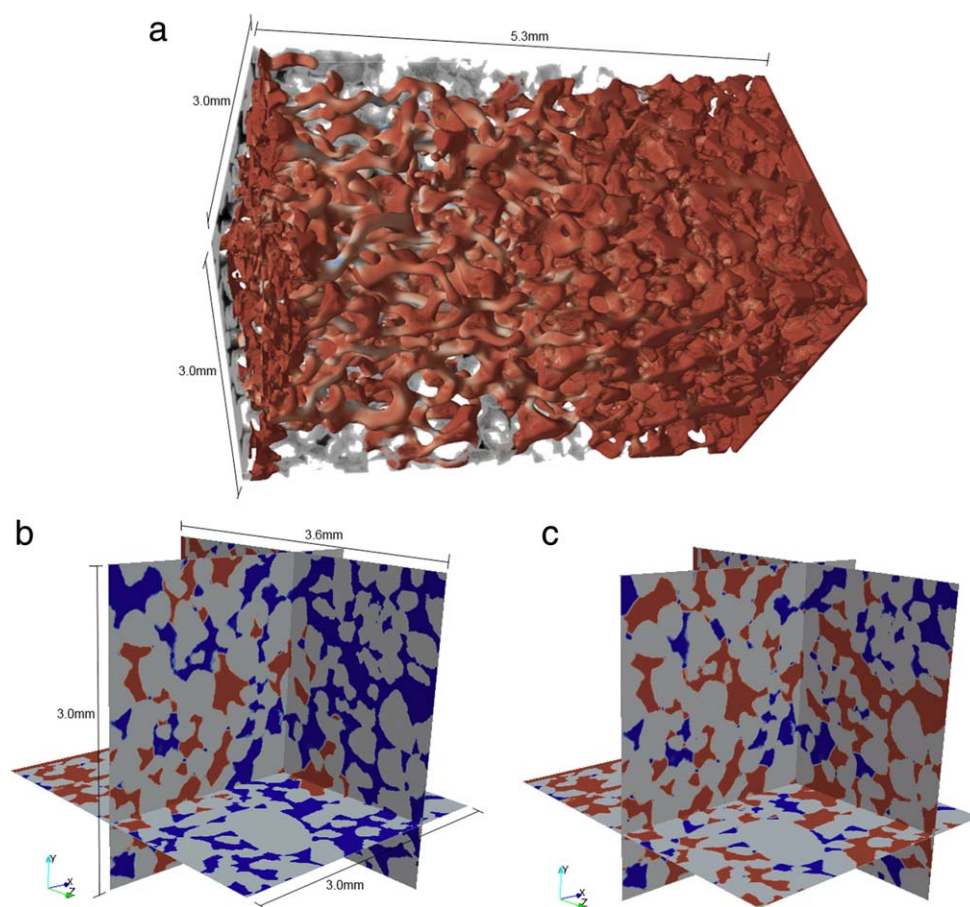


Figure 11. (a) Deduced 3-D pore space element with simulated desaturation (air phase: red isosurface) using the LB method. 2-D slices of wetting (blue) and nonwetting (red) phase distributions (b) at 50% saturation, and (c) at 23% saturation, respectively.

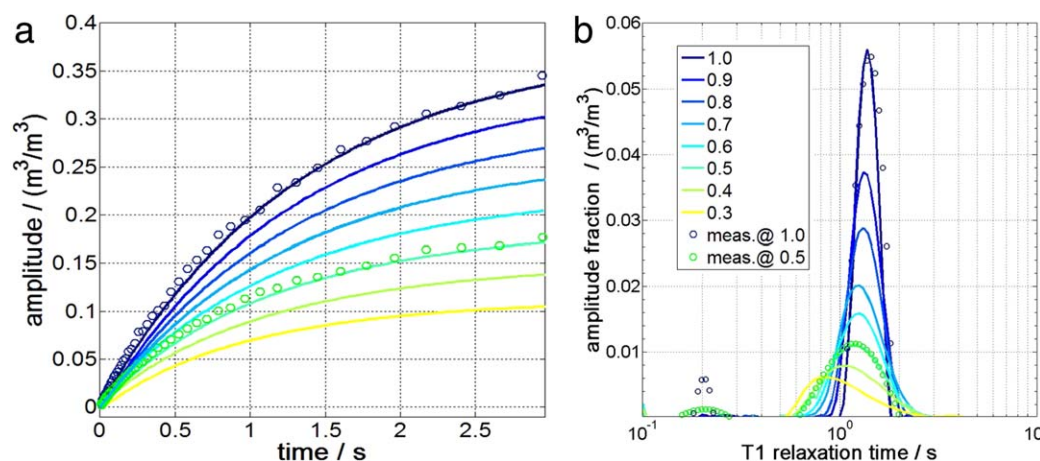


Figure 12. (a) Simulated (lines) and measured (circles) saturation recovery data for the FH31 sand sample and (b) inverted T_1 decay time distributions for the eight saturation degrees between 0.3 and 1.0.

of saturation compared to the laboratory data. Even though, when considering the different scales of the experiment and the CT image, a qualitative correlation can be observed.

Experimental NMR T_1 data were available for saturation states $S = 1$ and $S = 0.5$, respectively. At eight saturation levels ranging from $S = 0.9$ to $S = 0.3$, that were reached during the numerical drainage experiment (e.g., phase distributions at $S = 0.5$ and $S = 0.23$ in Figure 11), the NMR T_1 measurements were simulated and processed as discussed in section 3.2 (Figure 12a). In the NMR simulations, a numerical time step corresponded to 9.36×10^{-6} s of real time. As expected, the obtained NMR amplitudes correlated linearly with the degree of saturation. The simulated relaxation curves also correlated well with the experimental data available for full and 50% saturation.

Corresponding relaxation time distributions (Figure 12b) of the simulated NMR transients, which were derived using a regularized multiexponential fitting algorithm [Mohnke, 2010], were in very good agreement with the inversion results obtained for the respective measured NMR data. A second peak at low relaxation times that was prominent in the measured data was just slightly indicated in the simulated signals. This difference can probably be attributed to the limited resolution of the CT image (voxel size of $15 \mu\text{m}$), which does not capture tiny pores as well as rough surfaces (i.e., high local surface-to-volume ratios). Thus, they did not contribute to the simulated NMR relaxation signals.

Note that the relaxation time distribution at low saturation degrees (e.g., yellow line) was slightly shifted toward smaller times outside the initial distribution envelope (blue line) at full saturation, which reflects the pore size distribution of the sample. Such behavior that can be frequently observed in partially saturated rocks can possibly be attributed to residual water trapped in pore corners and films during desaturation that can exhibit very short relaxation times (cf. Figure 11b) [e.g., *Reservoir Technology*, 1996; *Thern*, 2014].

6. Conclusions

A numerical framework was developed that allows for the joint simulation of both the NMR relaxation and transport properties of partially saturated porous systems. Here, Lattice Boltzmann methods were used for jointly simulating two-phase flow and NMR T_1 relaxometry signatures in a steady state model environment. The models were validated by analytical solutions for simple spherical pore geometries. Moreover, numerical experiments were compiled using the complex 3-D pore space derived from micro-CT images. The presented numerical simulations agree well with laboratory observations regarding porosity, inner surface area, as well as NMR amplitudes, and decay times for varying water saturation degrees of a packed sand sample.

In this context, the numerical LB experiments can be used to investigate the observed complex behavior of NMR signatures of partially saturated porous media, such as rocks and soils. In that way, the actual spatial distribution of the pore fluids in a rock can be simulated and imaged jointly with the NMR relaxometry signatures. Consequently, such numerical experiments can be utilized to conduct numerical experiments

related to multiphase flow and NMR relaxometry signatures. For instance, the influence of hysteresis as well as of trapped water in desaturated pore corners on the corresponding NMR signals can be studied. Moreover, numerical experiments can be used to develop and to assess inverse modeling and interpretation schemes for NMR measurements on partially saturated porous systems, e.g., aiming to predict the transport properties.

Appendix A: Generalized Lattice Boltzmann Method

The technical information in this and the two following appendices gives an overview of the numerical method used in this work, and it is given for the sake of completeness, mainly summarizing Tölke *et al.* [2006]. Appendices A and B specify the LB collision operator. Subsequently, the Rothmann-Keller model extension for multiphase flow is described in Appendix C.

The matrix $\mathbf{M}$ given in Appendix B transforms the distributions $\mathbf{f}$ into moment space. The resulting moments are $\mathbf{m} = \mathbf{M}\mathbf{f}$. They are then relaxed according to $m_q^{p.c.} = m_q + s_q(m_q - m_q^{eq})$, which results in the post collision state of the moments, $m_q^{p.c.}$. s_q are the relaxation rates (not to be confused with NMR relaxation).

The equilibrium moments $\mathbf{m}^{eq}$ depend on the local state of the conserved quantities $\rho = m_0$, $\rho u_1 = m_3$, $\rho u_2 = m_5$, and $\rho u_3 = m_7$, and for the simulation of multiphase systems, additional terms were included for modeling the surface tension σ across the diffuse interface based on the gradient $\mathbf{C}$ of the phase field (the computation of $\mathbf{C}$ is described in Appendix C):

$$m_0^{eq} = m_0, \quad (A1a)$$

$$m_1^{eq} = (m_3^2 + m_5^2 + m_7^2) - \sigma \|\mathbf{C}\|, \quad (A1b)$$

$$m_3^{eq} = m_3, \quad m_5^{eq} = m_5, \quad m_7^{eq} = m_7, \quad (A1c)$$

$$m_9^{eq} = (2m_3^2 - m_5^2 - m_7^2) + \frac{1}{2} \sigma \|\mathbf{C}\| (2n_x^2 - n_y^2 - n_z^2), \quad (A1d)$$

$$m_{11}^{eq} = (m_5^2 - m_7^2) + \frac{1}{2} \sigma \|\mathbf{C}\| (n_y^2 - n_z^2), \quad (A1e)$$

$$m_{13}^{eq} = m_3 m_5 + \frac{1}{2} \sigma \|\mathbf{C}\| (n_x n_y), \quad (A1f)$$

$$m_{14}^{eq} = m_5 m_7 + \frac{1}{2} \sigma \|\mathbf{C}\| (n_y n_z), \quad (A1g)$$

$$m_{15}^{eq} = m_3 m_7 + \frac{1}{2} \sigma \|\mathbf{C}\| (n_x n_z), \quad (A1h)$$

$$m_2^{eq} = m_4^{eq} = m_6^{eq} = m_8^{eq} = m_{10}^{eq} = m_{16}^{eq} = m_{17}^{eq} = m_{18}^{eq} = 0. \quad (A1i)$$

The moments m_{q_c} , $q_c \in \{0, 3, 5, 7\}$ represent mass and momentum and are conserved during the collision. $n_{x,y,z}$ are the normal vector components in x , y , and z direction, respectively (cf. Appendix C). The nonzero relaxation rates are

$$s_1 = -s_e, \quad (A2a)$$

$$s_2 = -s_\xi, \quad (A2b)$$

$$s_4 = s_6 = s_8 = -s_q, \quad (A2c)$$

$$s_{10} = s_{12} = -s_\pi, \quad (A2d)$$

$$s_9 = s_{11} = s_{13} = s_{14} = s_{15} = -s_v, \quad (A2e)$$

$$s_{16} = s_{17} = s_{18} = -s_m. \quad (A2f)$$

The relaxation rate s_v is related to the kinematic viscosity ν by

$$s_v = \frac{2c^2 \Delta t}{6\nu + c^2 \Delta t}. \quad (A3)$$

The other relaxation rates s_e , s_ξ , s_q , s_π , and s_m can be freely chosen in the range $[0, 2]$ and may be tuned to improve accuracy and stability [Lallemand and Luo, 2000]. We applied "magic" parameters recommended for Stokes flow in [Ginzburg, 2006]:

$$s_e = s_\zeta = s_\pi = s_v, s_q = s_m = 8 \frac{2 - s_v}{8 - s_v}. \quad (\text{A4})$$

Appendix B: Transformation Matrix of Moments $\mathbf{M}$

The transformation matrix $\mathbf{M}$

$$M_{pq} = \Phi_{pq}, \quad p, q = 0, 1, \dots, 18 \quad (\text{B1})$$

is constructed from the orthogonal basis vectors

$$\Phi_{0q} = 1, \quad \Phi_{1q} = \overrightarrow{\xi}_q^2 - c^2, \quad \Phi_{2q} = 3(\overrightarrow{\xi}_q^2)^2 - 6\overrightarrow{\xi}_q^2 c^2 + c^4, \quad (\text{B2a})$$

$$\Phi_{3q} = \zeta_{q1}, \quad \Phi_{5q} = \zeta_{q2}, \quad \Phi_{7q} = \zeta_{q3}, \quad (\text{B2b})$$

$$\Phi_{4q} = (3\overrightarrow{\xi}_q^2 - 5c^2)\zeta_{q1}, \quad \Phi_{6q} = (3\overrightarrow{\xi}_q^2 - 5c^2)\zeta_{q2}, \quad (\text{B2c})$$

$$\Phi_{8q} = (3\overrightarrow{\xi}_q^2 - 5c^2)\zeta_{q3}, \quad (\text{B2d})$$

$$\Phi_{9q} = 3\zeta_{q1}^2 - \overrightarrow{\xi}_q^2, \quad \Phi_{11q} = \zeta_{q2}^2 - \zeta_{q3}^2, \quad (\text{B2e})$$

$$\Phi_{10q} = (2\overrightarrow{\xi}_q^2 - 3c^2)(3\zeta_{q1}^2 - \overrightarrow{\xi}_q^2), \quad (\text{B2f})$$

$$\Phi_{12q} = (2\overrightarrow{\xi}_q^2 - 3c^2)(\zeta_{q2}^2 - \zeta_{q3}^2), \quad (\text{B2g})$$

$$\Phi_{13q} = \zeta_{q1}\zeta_{q2}, \quad \Phi_{14q} = \zeta_{q2}\zeta_{q3}, \quad \Phi_{15q} = \zeta_{q1}\zeta_{q3}, \quad (\text{B2h})$$

$$\Phi_{16q} = (\zeta_{q2}^2 - \zeta_{q3}^2)\zeta_{q1}, \quad \Phi_{17q} = (\zeta_{q3}^2 - \zeta_{q1}^2)\zeta_{q2}, \quad (\text{B2i})$$

$$\Phi_{18q} = (\zeta_{q1}^2 - \zeta_{q2}^2)\zeta_{q3}, \quad (\text{B2j})$$

and the resulting matrix for D3Q19 becomes

$$\begin{bmatrix} 1. & (1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1) \\ c^2. & (-1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1) \\ c^4. & (1 & -2 & -2 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1) \\ c. & (0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 0 & 0) \\ c^3. & (0 & -2 & 2 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 0 & 0) \\ c. & (0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 & -1) \\ c^3. & (0 & 0 & 0 & -2 & 2 & 0 & 0 & 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 & -1) \\ c. & (0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & -1) \\ c^3. & (0 & 0 & 0 & 0 & 0 & -2 & 2 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & -1) \\ c^2. & (0 & 2 & 2 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -2 & -2 & -2) \\ c^4. & (0 & -2 & -2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -2 & -2 & -2) \\ c^2. & (0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 0 & 0) \\ c^4. & (0 & 0 & 0 & -1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 0 & 0) \\ c^2. & (0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0) \\ c^2. & (0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1) \\ c^2. & (0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 0 & 0) \\ c^3. & (0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & 0 & 0) \\ c^3. & (0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1) \\ c^3. & (0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & -1 & -1) \end{bmatrix}.$$

Appendix C: Rothmann-Keller Multiphase Model

Two dimensionless density fields ρ_A and ρ_B representing the wetting and nonwetting phase, respectively, are introduced to define an order parameter ϕ

$$\phi = \frac{\rho_A - \rho_B}{\rho_A + \rho_B}, \quad (C1)$$

such that $\phi = 1$ signifies the wetting phase and $\phi = -1$ signifies the nonwetting phase. The value of ϕ is constant in both bulk phases and varies only in the diffuse phase interface.

The density fields ρ_A and ρ_B are advected with the following LB models for $p \in \{A, B\}$ and $q \in \{0, \dots, 18\}$:

$$g_{p,q}(t + \Delta t, \vec{x} + \vec{\xi}_q \Delta t) = g_{p,q}^{eq}(\rho_p(t, \vec{x}), \vec{u}(t, \vec{x})), \quad (C2)$$

where $\rho_p = \sum_{q=0}^{18} g_{p,q}$ and $\vec{\xi}_q$ are defined as in equation (12) described in section 3. The equilibrium distribution function $g_{p,q}^{eq}$ is given according to [Kehrwald, 2002]:

$$g_{p,q}^{eq}(\rho_p, \vec{u}) = \omega_q \rho_p \left(1 + \frac{3}{c^2} \vec{\xi}_q \cdot \vec{u} \right), \quad (C3)$$

where the weights are $\omega_0 = 1/3$, $\omega_q = 1/18$ for $q \in \{1, \dots, 6\}$, and $\omega_q = 1/36$ for $q \in \{7, \dots, 18\}$. Equations (C2) and (C3) result in an advection diffusion equation for ρ_p with a constant diffusion of $D = \frac{1}{6} c^2 \Delta t$, which means the phases are mixing into each other. To reverse that unwanted diffusion, a demixing algorithm can be applied; this is often referred to as “recoloring algorithm” [Kehrwald, 2002]. We chose the recoloring algorithm given in Tölke [2002].

To take the surface tension into account (cf. Appendix A), the gradient of ϕ has to be computed. This is done by

$$\mathbf{C}(t, \vec{x}) = \frac{3}{c^2 \Delta t} \sum_{q=0}^{19} \omega_q \vec{\xi}_q \phi(t, \mathbf{x} + \vec{\xi}_q \Delta t). \quad (C4)$$

To compute the normal vector on the liquid-gas interface, $\mathbf{C}$ is normalized as follows:

$$n_\alpha = \frac{C_\alpha}{\|\mathbf{C}\|}, \quad \alpha \in \{x, y, z\}. \quad (C5)$$

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